Luis Enrique Dalmau Coopat

Full-Stack AI Engineer · Computer Vision · YOLOv9

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Professional Summary

Full-Stack AI Engineer with 3+ years building production systems across computer vision, ML pipelines, and full-stack web applications. Author of a proprietary computer-vision library used to train and deploy YOLOv9 object-detection models for industrial IIoT predictive maintenance. B.S. in Computer Science, hands-on MLOps, and a teaching background in AI, ML, and distributed systems. Spanish native, English B2 (professional working proficiency).

Skills

AI & Computer Vision: Python · PyTorch · TensorFlow · OpenCV · Deep Learning · MLOps · Object Detection · Image Segmentation · Auto-labeling (GroundingDINO, SAM, Florence-2) · Synthetic Data (Stable Diffusion) · Albumentations · TensorBoard · scikit-learn · pandas · NumPy
Backend: FastAPI · SQLAlchemy 2.0 · Alembic · Django · Pydantic · REST APIs · JWT/OAuth2 · APScheduler
Frontend: React 18 · Next.js 14 · TypeScript · Tailwind CSS · TanStack Query · Formik · Chart.js
Databases: PostgreSQL · MongoDB · MySQL · Redis
DevOps & Cloud: Docker · Docker Compose · Heroku · AWS S3 · Nginx · Git · GitLab CI · Linux
Testing & Quality: pytest · pytest-asyncio · pytest-cov · moto · httpx · ESLint · Prettier

Experience

AI Engineer — Computer Vision
SORBA.ai (Sorbotics LLC) · Jacksonville, USA (Remote) — via Anukys, Málaga, Spain (since 10/2025) | 01/2024 – Present

- Built and optimized CV training pipelines covering 10+ architectures (YOLOv5/v9, EfficientDet, MobileNet v2, RetinaNet, Faster R-CNN, Mask R-CNN, CenterNet, SSD).
- Shipped these capabilities across 22 company projects, from dataset preparation to on-plant deployment.
- Integrated synthetic data generation (Stable Diffusion, GAN, VAE), auto-labeling (GroundingDINO, Florence-2), and model optimization (TensorRT, TFLite).
- Built real-time video processing for industrial detection: fire/smoke, gas leaks, PPE, gauge OCR, license plates, and spatial tracking.
- Packaged ML models for Debian/RHEL. Managed GPU resources for concurrent training.

Full-Stack Developer (Backend focus)
Aerial Underground Construction Inc. · Kissimmee, FL, USA (Remote, concurrent with SORBA.ai) | 09/2024 – 10/2025

- Built logistics platform backend (FastAPI + SQLAlchemy + Alembic + PostgreSQL): JWT auth, RBAC (3 roles), 20+ REST endpoints.
- Implemented commissions, daily/weekly/monthly reports, PDF/Excel invoices (ReportLab/openpyxl), AWS S3 (boto3), scheduled jobs (APScheduler).
- Developed ~35% of the frontend codebase (Next.js 14 + React + TypeScript) with automated tests and Docker/Heroku deployment.

Assistant Professor
University of Havana · Faculty of Mathematics & Computer Science (part-time, concurrent) | 01/2023 – 01/2025

- Taught Computer Networks, Machine Learning, AI, Operating Systems, and Distributed Systems; led practicals, graded exams, mentored students.

Projects

Computer Vision Library (proprietary) — author: YOLOv9 training, auto-labeling, synthetic data, GPU management.
Logistics Management System — FastAPI + Next.js + PostgreSQL; RBAC, reports, invoices, AWS S3.
YOLO-3D — Real-time 3D object detection: YOLOv11 + Depth Anything v2.
Image Analyzer (BLIP) — Multimodal image analysis: descriptions, keywords.
LicitAI — FastAPI + React + MongoDB + Google Gemini (LLM) + Stripe.
Protectora RRSS — Pet-adoption diffusion: FastAPI + React + Telegram bot + SMTP.
ML Trainer Engine — CV training with SOTA models (RF-DETR, EfficientDet, MobileNet, DeepLabV3).

Education & Certifications

B.S. in Computer Science — University of Havana (2017–2022). Thesis: "Graphic generation of signs of the Cuban Sign Language".
CS50's Introduction to AI with Python — HarvardX (edX) · **Data Science & Machine Learning** — Sololearn.

Languages

Spanish: Native · **English:** B2 (Professional Working Proficiency)